RANSBET IIMS

Intelligent Inventory Management System
Explained Simply — A Complete Beginner's Guide

A final-year project for Ransbet Supermarket, Tarkwa

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Live system: https://aizen004.pythonanywhere.com

1. The Story: Why We Built This

Imagine a busy supermarket called **Ransbet**. Every day hundreds of items are bought and sold. For a long time, the staff wrote down their stock in notebooks and spreadsheets by hand.

That caused problems. Popular items ran out and customers left unhappy. Other items were over-ordered and sat on the shelf wasting money. Nobody could tell what would sell next week, and mistakes or losses were only noticed days later.

Our job was to build a smart computer helper that fixes all of this — one that always knows what is in stock, warns the staff early, and even **predicts the future**.

2. What We Built (in one breath)

We built a **website** that the supermarket staff log into. It:

- keeps a live list of every product and how many are left;
- shows everything on a colourful dashboard with charts;
- warns when an item is about to run out, and suggests how much to reorder;
- predicts how much of each product will sell over the next month;
- spots strange events (like a sudden unusual drop in stock);
- can be opened from any phone or computer, anywhere, with a password.

3. The Big Picture: Three Parts, Like a Restaurant

Our system is made of three parts that work together. The easiest way to picture it is a restaurant:

- **The dining room (what you see)** — the screens, buttons and charts. In our system this is the **dashboard** the staff look at.
- **The kitchen (does the work)** — where the cooking and thinking happens. In our system this is the **program** running on the server. It checks passwords, does the calculations, and decides what to show.
- **The storeroom (remembers everything)** — where all the ingredients are kept. In our system this is the **database**, which remembers every product and sale.

The dining room never goes into the storeroom directly — it always asks the kitchen. That keeps everything tidy and safe. Programmers call this a **three-tier architecture**.

4. The Tools We Used (and what each one does)

Tool	In plain words
Python	The main language we wrote our instructions in.

Flask	A helper that lets Python run a website (the 'kitchen').
React	Builds the pretty, moving dashboard (the 'dining room').
MySQL	The filing cabinet that remembers all the data (the 'storeroom').
Prophet	A 'fortune teller' that predicts future sales from the past.
Isolation Forest	A 'detective' that spots unusual, suspicious events.
Git & GitHub	A time machine and online backup for all our code.
VS Code	The desk (program) where we actually write the code.
PythonAnywhere	The place on the internet where the website lives 24/7.

5. How the Parts Talk to Each Other

When you open the dashboard, here is what happens, step by step:

- Your screen asks the program: 'Who is logged in, and what are today's numbers?'
- The program checks your password is still valid, then looks inside the database.
- The database hands back the answers (how many products, today's sales, and so on).
- The program sends those answers to your screen as a tidy message.
- Your screen draws the cards and charts from that message.
- It quietly asks again every 5 seconds — that is why the numbers always look up to date, even if someone else made a change.

6. Everything the System Can Do

- **Product list** — add, edit and remove products.
- **Live stock** — see exactly how many of each item are left, with traffic-light colours (green = fine, yellow = getting low, red = reorder now).
- **Forecasting** — predict how much will sell in the next 30 days.
- **Anomaly alerts** — flag unusual sales or stock changes.
- **Restock alerts** — say which items to reorder and how many.
- **Dashboard** — charts and numbers that tell the whole story at a glance.
- **Barcode scanning** — scan a product instead of typing it.
- **Reports** — download neat PDF or spreadsheet reports.
- **Logins & roles** — different people get different permissions.
- **History log** — every action is recorded with who did it and when.

7. The Clever Bit: Predicting the Future

How do weather forecasters guess tomorrow's weather? They study what happened on many days before and find patterns. Our system does the same with **sales**.

We gave it about **two years** of daily sales. It noticed patterns by itself — weekends are busier than weekdays, December is very busy, public holidays cause spikes — and learned how each product normally behaves. Then it makes a smart guess for the next 30 days for every product.

To check the guesses are good, we played a fair game: we hid the most recent month of real sales, asked the system to guess it, and compared the guess to what actually happened. On average it was about **87% accurate** — better than the project target of 85%.

8. The Watchful Eye: Spotting Strange Things

Most days, a product sells a fairly normal amount. But sometimes something odd happens: ten times the usual amount disappears, or sales suddenly drop to zero for no clear reason. That could be a typing mistake, theft, a damaged batch, or a surprise rush of customers.

Our 'detective' learns what **normal** looks like for each product. When a day does not fit the normal pattern, it raises a flag so the manager can look into it. This helps catch losses and errors early instead of weeks later.

9. Where Everything Is Remembered: The Database

Think of the database as a set of labelled boxes. Each box (we call it a **table**) holds rows, like lines in a notebook. The boxes are linked, so a sale always knows which product it belongs to. Our main boxes are:

- **Products** — every item, its price and how many are in stock.
- **Sales** — every sale (this is what the fortune teller studies).
- **Suppliers** and **Categories** — who provides items and how they are grouped.
- **Stock movements** — a logbook of every stock change.
- **Forecasts** and **Anomaly flags** — the predictions and the warnings.
- **Users** — the login accounts.
- **History log** — who did what, and when.

10. Different Keys for Different People

Not everyone should be able to do everything, so we gave out three kinds of 'keys':

- **Store Manager** — the master key. Sees forecasts, approves reorders, can delete things.

- **Inventory Staff** — can add products, record sales, and receive or adjust stock, but cannot delete or change settings.
- **Business Owner** — mainly looks at reports and trends.

Before the system lets anyone do something, it checks their key. This is called **role-based access control**.

11. Keeping It Safe

- Passwords are **scrambled** before they are saved, so even we cannot read them.
- The system checks your role on the server for every action, so the rules cannot be tricked from outside.
- Forms are protected so a malicious website cannot make your browser do something without permission.

12. Putting It On the Internet

On our own laptops, the system only worked for us. To let anyone use it, we copied it to a free online service called **PythonAnywhere**. It keeps the website running day and night on its own computers, so the link works for everyone in the world — **even when our laptops are switched off**.

You can try it now: **https://aizen004.pythonanywhere.com** (sign in with manager@ransbet.com / manager123).

13. How To Switch It On (on our laptop)

- Double-click **START_RANSBET.bat** — this starts the database and the website.
- Open a web browser and go to **http://127.0.0.1:5000**.
- Log in and use it, exactly like the live version.

14. Words You Might Hear, Explained

Word	Simple meaning
Server	A computer that runs the website and serves it to others.
Database	An organised store of information (our 'storeroom').
Frontend	The part you see and click (the 'dining room').
Backend	The hidden part that does the work (the 'kitchen').
API	The messenger that carries questions and answers between them.
Machine learning	Teaching a computer to find patterns and make guesses.

Forecast	An educated guess about the future, based on the past.
Anomaly	Something unusual that does not fit the normal pattern.
Deploy	To put the finished system online for people to use.
Real-time	Updating by itself, almost instantly, without reloading.

In one sentence: *we built a smart website that watches the supermarket's stock, warns the staff early, predicts the future from the past, and can be used by anyone, anywhere — and we understand every part of how it works.*